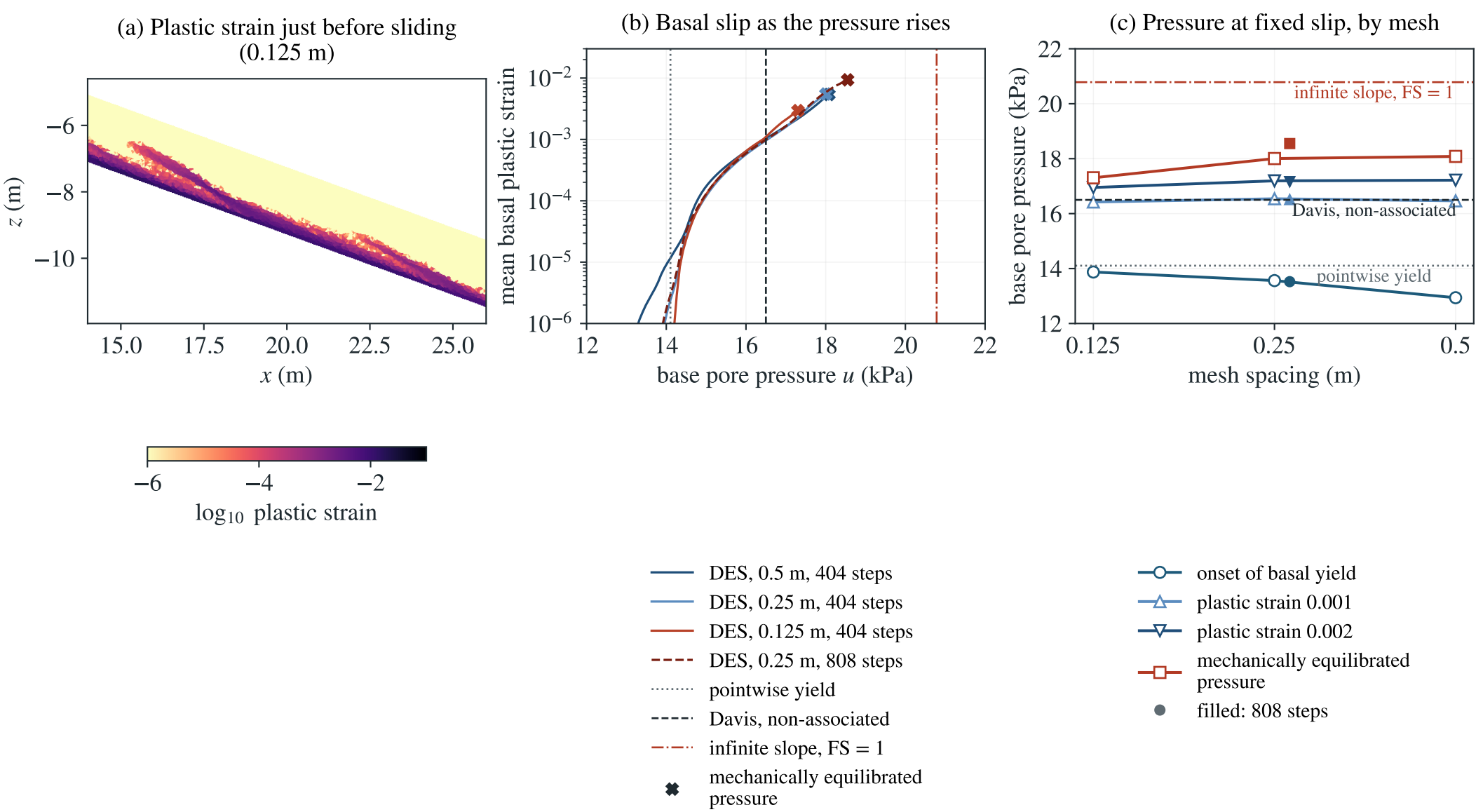
(a) Plastic strain just before sliding (0.125 m)
z (m)
−6
−8
−10
15.0
17.5
20.0
22.5
25.0
x (m)
−6
−4
−2
log10 plastic strain
(b) Basal slip as the pressure rises
mean basal plastic strain
10−2
10−3
10−4
10−5
10−6
12
14
16
18
20
22
base pore pressure u (kPa)
(c) Pressure at fixed slip, by mesh
base pore pressure (kPa)
22
20
18
16
14
12
infinite slope, FS = 1
Davis, non-associated
pointwise yield
0.125
0.25
0.5
mesh spacing (m)
DES, 0.5 m, 404 steps
DES, 0.25 m, 404 steps
DES, 0.125 m, 404 steps
DES, 0.25 m, 808 steps
pointwise yield
Davis, non-associated
infinite slope, FS = 1
mechanically equilibrated pressure
onset of basal yield
plastic strain 0.001
plastic strain 0.002
mechanically equilibrated pressure
filled: 808 steps